

Same Scene, Different Story: Evaluating Robot Perception Across Scene Phases in the Wild

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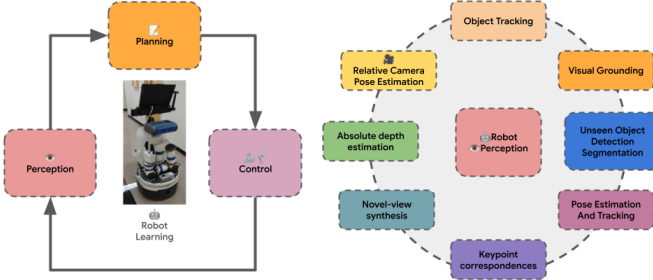


Fig. 1: **RPX**: a human-carried rig (D435 + T265) captures 100 scenes in three phases (Clutter → Interaction → Clean); a head-mounted GoPro records in parallel during Interaction. Seven tasks are evaluated on the same scenes; Φ (State-Transition Robustness) and $\mathcal{J}$ (JEDI) quantify deployment readiness.

Abstract—Pretrained perception models—depth estimators, trackers, VLMs—now serve at every level of the robot learning stack: as preprocessing in modular pipelines, as frozen backbones in robot foundation models, and as visual reasoning modules in agentic systems. Yet these models are selected from internet benchmarks and deployed on robot hardware without re-evaluation. We present **RPX** (Robot Perception X), a unified benchmark that evaluates pretrained perception under robotic scene conditions. 100 real-world scenes are captured in three phases—Clutter, Interaction, Clean—on commodity RGB-D sensors; seven tasks are evaluated with up to 10 models each, all zero-shot. Φ (repeated-measures MANOVA) tests whether a model’s multi-metric vector shifts across scene states; $\mathcal{J}$ (geometric-mean desirability) gives one desirability index where any failing metric drags it toward zero. **RPX** replaces guesswork with evidence for perception backbone selection in downstream robot tasks. Evaluating 29 models across depth and pose estimation, we find that phase change significantly degrades every depth model ($\Phi \leq 0.79$, $p < 0.001$) but not pose estimators ($p > 0.05$), that near-range depth collapses 2–4 $\times$ during interaction, and that model rankings invert across tasks—no single foundation model dominates. Code and data: <https://anonymous.4open.science/r/RPX>.

I. INTRODUCTION

Pretrained perception models—depth estimators, trackers, pose regressors, vision-language models—now underpin every layer of the robot learning stack. Modular pipelines chain them as preprocessing [1], [2]; robot foundation models inherit them as frozen backbones [3], [4], [5]—the single largest factor in VLA success [6]; agentic systems invoke them repeatedly

in closed-loop reasoning. Yet the choice of *which* model to deploy is made by convention, not by evidence: the depth model is ranked on NYU [7], the tracker on DAVIS, the VLM on an internet QA benchmark—each a different camera, different scenes, and none tests scenes that change during manipulation.

Three structural gaps explain why leaderboard rankings fail to predict deployment performance: (i) *sensor mismatch*—models trained on internet images are deployed on commodity RGB-D sensors with different resolution, noise, and field of view; (ii) *task silos*—each perception task is benchmarked in isolation, so a practitioner cannot compare candidates across the tasks a single pipeline requires; (iii) *static scenes*—existing benchmarks freeze the scene, but robots operate in environments that change during manipulation, demonstration, and collaboration. No existing benchmark addresses all three (Table I).

We present **RPX** (Robot Perception X), a unified benchmark that evaluates pretrained perception under robotic scene conditions. 100 real-world scenes are each captured in three phases—**Clutter**, **Interaction**, **Clean**—on a D435+T265 rig; a head-mounted GoPro captures an egocentric view during Interaction. Seven tasks (image depth, video depth, tracking, relative pose, and three levels of visual grounding) are evaluated with up to 10 models each, all zero-shot on the same scenes. Two complementary diagnostics replace single-number leaderboards: Φ (repeated-measures MANOVA) tests whether a model’s multi-metric profile shifts across scene phases; $\mathcal{J}_{\min}$ (geometric-mean desirability) gives one desirability index where any failing metric drags it toward zero. The `rpx-benchmark` toolkit is open-source; adding a model requires one adapter file.

Evaluating 29 models across depth and pose estimation, we find that phase change significantly degrades every depth model ($p < 0.001$) but not pose estimators ($p > 0.05$), that near-range depth collapses 2–4 $\times$ during interaction, and that model rankings invert across tasks—no single foundation model dominates.

II. RELATED WORK

Foundation models as robot perception backbones. Pretrained models now serve as frozen encoders in VLAs [4], [3], [5], as preprocessing modules in modular pipelines [1], [2], and as pseudo-GT generators for world models [14]. Li et al. [6] showed backbone choice is the single largest factor in VLA success (in simulation); adapting perception to deployment via interactive teaching [15] directly improves

TABLE I: Comparison with benchmarks relevant to robot perception. ✓ = supported; ~ = partial; ✗ = not supported. ||Policy-learning dataset. *Static extrinsics only. ‡Trajectories/trials.

	Perception					Robot Data		
	RPX	ScanNet [8]	YCB-V [9]	BOP [10]	RoboDepth [11]	DROID [12]	RH20T [13]	OK-Robot [1]
<i>Data properties</i>								
Real-world RGB-D	✓	✓	✓	✓	✗	~	✓	✓
Metric depth GT	✓	✓	✓	✓	✓	~	✓	✓
6-DoF camera pose	✓	✓	✗	✗	✓	~*	~*	✗
Instance masks	✓	✓	✓	✓	✓	✗	✗	✗
Multi-task GT on shared scenes	✓	~	✗	✗	✗	✗	~	✗
<i>Protocol</i>								
Three-phase capture	✓	✗	✗	✗	✗	✗	✗	✗
Human interaction phase	✓	✗	✗	✗	✗	✓	✓	✗
Shared objects across phases	✓	✗	✗	✗	✗	✗	✗	✗
Scene difficulty strata	✓	✗	✗	✗	✓	✗	✗	✗
<i>Evaluation methodology</i>								
Φ (phase + ego-exo)	✓	✗	✗	✗	✗	✗	✗	✗
$\mathcal{J}$ (desirability index)	✓	✗	✗	✗	✗	✗	✗	✗
# Scenes	100	1.5K	92	7	n/a	564	147	10
# Frames	75K	2.5M	134K	var.	n/a	76K [‡]	110K [‡]	171 [‡]
Focus	deploy.	recon.	pose	pose	depth	imit.	imit.	manip.

manipulation on real hardware. Despite this evidence, no existing benchmark evaluates these models collectively on robot-grade data.

Per-task perception benchmarks. Each perception capability has its own benchmark ecosystem: NYU Depth V2 [7] for depth, COCO [16] for detection, KITTI [17] for odometry and depth—each a different camera, different scenes, different domain. RoboDepth [11] tests depth robustness but applies synthetic corruptions (blur, noise, weather) to existing benchmark images; the perturbations are post-hoc, not arising from natural scene dynamics. SenseShift6D [18] sweeps camera exposure/gain settings for 6-DoF pose, but is single-task with static scenes. These benchmarks serve their respective communities well, but a model selected by NYU-AbsRel rank cannot be checked against the same scenes a tracker is validated on.

Robot-centric datasets. Table I compares RPX to related benchmarks along axes relevant to perception backbone selection for robotics. YCB-Video [9] and BOP [10] provide object pose GT but use static cameras in controlled setups—no scene-state variation. DROID [12] and RH20T [13] collect large-scale real manipulation trajectories but provide no per-frame perception GT (no depth error, no tracking metrics). OK-Robot [1] demonstrates real-world deployment but evaluates the full pipeline, not individual perception modules. The gap is clear: no existing benchmark provides multi-task perception evaluation on shared real-world scenes with controlled scene-state variation and robot-grade metric GT.

Robustness evaluation. Vision robustness benchmarks [19], [20] apply synthetic perturbations to measure fragility. RPX differs in that the variation is natural: a human rearranges objects, creating the occlusion, clutter change, and hand presence that define real manipulation episodes. Difficulty is stratified by annotation-refinement effort, not corruption severity (§III-C).

III. BENCHMARK DESIGN

A. Three-Phase Protocol

Each of 100 scenes is recorded three times with a hand-carried RealSense D435 (RGB-D, 640×480) paired with a T265 (6-DoF VIO): **Clutter** ($\mathcal{P}_0$)—the scene as found; **Interaction** ($\mathcal{P}_1$)—objects rearranged by hand, mirroring learning-from-human data [15], [12]; **Clean** ($\mathcal{P}_2$)—the reorganised scene. A head-mounted GoPro (1920×1080, RGB-only) records in parallel during Interaction for ego-vs-exo comparison. Background, objects, lighting, and sensors are fixed; only scene state changes. Each phase yields ~250 frames (~75K total).

Human-carried capture provides a conservative motion stress test—generally noisier than wheeled or legged platforms—and scales to 100 diverse environments (homes, offices, kitchens, outdoor) that would be difficult to access with current robot hardware. The viewpoint height (~1.0–1.5 m) overlaps with the operating envelopes of humanoid and mobile manipulator platforms, though it does not replicate the motion profile of any specific robot.

B. Evaluation Primitives

Each task produces K metrics per (scene, phase) cell. We compute two quantities per model.

State-Transition Robustness (Φ). Does this model care that we moved things around? We run a one-way repeated-measures MANOVA [21] with scenes ($S=100$) as subjects, phase ($P=3$) as the within-subjects factor, and the K -metric vector (z-scored, direction-aligned) as the response. Wilks’ $\Lambda = |E|/|H+E|$ measures the ratio of within-phase to total scatter; we convert to an effect size:

$$\Phi = \Lambda^{1/r}, \quad r = \min(K, P-1) = 2 \quad (1)$$

$\Phi \approx 1$: negligible phase effect. $\Phi \leq 0.86$: large effect (Cohen [22]). Significance is assessed via Rao’s F ; per-transition decomposition uses Hotelling T^2 (Holm-corrected). Because the test is multivariate, a model cannot mask degradation on one metric behind stability on another.

JEDI ($\mathcal{J}_{\min}$). Is this model actually good? Each metric k is mapped to a desirability $d_k \in [0, 1]$ using frozen, direction-aware bounds. The composite is the geometric mean:

$$\mathcal{J} = \left(\prod_{k=1}^K d_k \right)^{1/K}, \quad \mathcal{J}_{\min} = \min_p \mathcal{J}_p \quad (2)$$

One catastrophic metric drags $\mathcal{J}$ to zero. $\mathcal{J}_{\min}$ (worst-phase desirability) is the deployment ranking: it rewards models that are both accurate *and* stable. Φ and $\mathcal{J}$ are orthogonal—a uniformly poor model is trivially phase-invariant ($\Phi \approx 1$, $\mathcal{J} \approx 0$).

C. Difficulty Stratification (ESD)

Scene difficulty is derived from 31 features across 10 categories (annotation effort, occlusion, depth quality, camera motion, scene geometry, lighting, etc.). The anchor signal is *annotation-refinement effort*—the number of human correction



Fig. 2: Capture setup. *Top-left*: D435+T265 rig on laptop mount. *Top-right*: sensor detail. *Bottom*: three phases of one scene; GoPro visible during Interaction.

TABLE II: Seven perception tasks on the same 100 scenes. *Ego*: task also evaluated on the GoPro egocentric stream. *K*: metrics in the MANOVA.

#	Task	Models (top to 10)	Ego	K
T1	Image Depth	DA3 [24], DA-V2 [25], Depth Pro [26], UniDepth V2 [27], Metric3D V2 [28], MoGe-2 [29], Lotso-2 [30], HyDen, ZipDepth, FE2E	✓	4
T2	Video Depth	VGGT [31], DA3 [24], VGGT [32], GemDepth [33], MoSTIR [34], ChronoDepth [35], MoSTIR [36], Video-DA [37], DepthCrafter [38], RollingDepth [39]	✓	6
T3	Tracking	SAM 3.1, SAM 2 [23], SAM 2++, SAMURAI [40], EdgeTAM [41], DeAOT [42], Cutie [43], MOTIP, MASA, Grounded-SAM2	✓	4
T4	Rel. Pose	DA3 [24], Reloc3R [44], VGGT [31], PG3X [45], MASOR [46], CUT3R [47], Fast3R [48], MUS3R [49], DUS3R [50], MoSTIR [36]	✓	3
T5	General VQA	PaliGemma 2 [51] (3B, 10B), Gemma 4 [52] (4B, 12B), Qwen2.5-VL [53] (3B, 7B), InternVL 2.5, LLaVA-OV, Phi-3.5-Vision [54], Idefic3 [55]	✓	—
T6	Spatial VQA	(same 10 VLMs as T5)	✓	—
T7	In-Context VQA	(same 10 VLMs as T5)	✓	—

iterations to produce accurate masks—as a proxy for perceptual ambiguity. Tertile cuts yield Easy / Medium / Hard splits (33/33/34 scenes); all phases share the same tier.

D. Dataset

100 indoor/outdoor scenes, ~ 70 object categories, $\sim 75,000$ frames. Per frame: RGB, metric depth (D435), 6-DoF pose (T265 VIO, validated at $1.53^\circ / 4.68\text{cm}$ median via ARTag-board references across 70 scenes), instance masks and tracklets (SAM2 + GroundingDINO + human refinement), egocentric video (GoPro), crowd-sourced object descriptions. The model-assisted annotation pipeline follows the same paradigm as SA-V/SA-1B [23] and BURST: GroundingDINO proposes keyframe detections, SAM2 propagates masks across frames, and human annotators correct errors on flagged keyframes. All masks have been manually verified; the human corrections decouple GT quality from any single model’s predictions. Released under CC BY 4.0.

IV. TASKS & MODELS

RPX frames the benchmark as seven perception decisions (Table II), all evaluated zero-shot on the same 100 scenes. Φ and $\mathcal{J}$ are computed on the canonical metric sets from the dominant benchmark in each field.

T1–T2 (Depth) are evaluated in two modes: *Image* (single RGB frame) and *Video* (full phase clip); both share the same GT and per-frame accuracy metrics. Video depth adds temporal consistency metrics (TGM, TGSE). DA3 appears in

both, yielding a clean frame-vs-video ablation. **T3** (Tracking) initialises with GT first-frame masks; the Interaction phase stress-tests occlusion robustness. **T4** (Relative Pose) uses $\sim 60\text{K}$ deterministic image pairs; Metric AUC jointly thresholds rotation and metric-scale translation. **T5–T7** (VQA) evaluate three levels of visual grounding. All VQA tasks require bounding-box outputs rather than free-text answers, evaluated via IoU—eliminating text-matching bias and unifying grounding with question answering. T5 (General) tests object recognition and attribute grounding. T6 (Spatial) tests depth-grounded spatial reasoning (ordering, distance, relative 3D position). T7 (In-Context) presents multi-image context requiring cross-frame reasoning. T3, T5–T7 are additionally evaluated on the egocentric GoPro stream during Interaction (GT masks and tracklets available for both streams), enabling a direct ego-vs-exo comparison at $6.8\times$ resolution.

Toolkit. `rpX-benchmark` is pip-installable (MIT); adding a model requires one adapter file (*bring your own model*).

V. EXPERIMENTS

All models are evaluated zero-shot on $100\text{ scenes} \times 3$ phases (depth tasks) or 2 phases (RCPE: clutter vs. clean, since cross-phase relative pose is undefined across separate captures). Φ and $\mathcal{J}$ are computed on the complete-case subset ($n = 100$ scenes per cell).

A. Overview: Does Phase Matter?

Table III presents the headline result. For both depth tasks (T1, T2), Φ is significant for every model ($p < 0.001$, Holm-corrected), with all 20 models showing large effect sizes ($\Phi \leq 0.86$). The Clutter→Interaction transition dominates for 15/20 depth models. Relative pose estimation (T4) tells a different story: none of the 9 complete models reaches significance ($p > 0.05$ on all splits), and Φ ranges from 0.81–0.99. Phase change measurably degrades spatial perception (depth) but not geometric estimation (pose)—suggesting that the mechanisms are distinct.

B. T1: Image Depth

All 10 image depth models show large, significant phase effects ($\Phi \leq 0.79$, $p < 0.001$; Table IV). Metric3D V2 achieves the highest $\mathcal{J}_{\min}$ (0.71) with near-zero spread ($\mathcal{J}_{\Delta} = 0.01$)—uniformly good across phases—yet has the highest Φ among metric models (0.68): the most accurate model is also the most phase-sensitive. DA3 Metric-L is the most affected ($\Phi = 0.53$, lowest per-transition $\Phi_{0 \leftrightarrow 1} = 0.23$) but retains $\mathcal{J}_{\min} = 0.70$. Among relative models, FE2E is a notable outlier: $\mathcal{J}_{\min} = 0.65$ rivals the metric group despite requiring per-scene alignment.

C. T2: Video Depth

VGGT dominates: highest $\mathcal{J}_{\min}$ (0.70), highest Φ (0.77), and the only model whose TGM is phase-invariant (Table V). DA3 (video) achieves comparable $\mathcal{J}_{\text{overall}}$ (0.72) but drops sharply during Interaction ($\mathcal{J}_{\Delta} = 0.15$, $\Phi_{0 \leftrightarrow 1} = 0.23$). The bottom three models (Video-DA, DepthCrafter, RollingDepth)

TABLE III: Phase-robustness summary. Best/worst model per task ranked by $\mathcal{J}_{\min}$. Φ : omnibus phase-robustness ($\Phi \leq 0.86$ = large effect). T4 uses 2 phases (clutter vs. clean); depth tasks use 3. [†]MASt3R pending on 2/3 difficulty splits.

Task	K	M	Best model	$\mathcal{J}_{\min} \uparrow$	Worst model	$\mathcal{J}_{\min}$	Φ (range)	Hardest transition	p
T1: Image Depth	4	10	Metric3D V2	0.711	Lotus-2	0.351	0.53–0.79	Clu→Int	<.001
T2: Video Depth	6	10	VGGT	0.700	RollingDepth	0.203	0.45–0.77	Clu→Int	<.001
T3: Tracking	4	10				<i>in progress</i>			
T4: Rel. Pose	3	9 [†]	DA3	0.380	MonST3R	0.058	0.81–0.99	Clu→Cln	>.05
T5: General VQA	–	10				<i>in progress</i>			
T6: Spatial VQA	–	10				<i>in progress</i>			
T7: In-Context VQA	–	10				<i>in progress</i>			

TABLE IV: T1: Image depth ($K=4$: AbsRel, RMSE, SILog, δ_1). $\mathcal{J}$: per-phase desirability (geometric mean). $\mathcal{J}_{\min}$: worst-phase. $\mathcal{J}_{\Delta}$: spread. Phase-transition Φ : pairwise phase-robustness. Scene-tier Φ : robustness within ESD difficulty strata. η^2 : difficulty amplification of the phase effect. [‡]Alignment regime under review.

Model	Venue	Out	Params	Desirability $\mathcal{J} \uparrow$				Summary		Overall	Phase-transition $\Phi \uparrow$			Scene-tier $\Phi \uparrow$			Amplif.
				overall	clu	int	cln	$\mathcal{J}_{\min} \uparrow$	$\mathcal{J}_{\Delta} \downarrow$	$\Phi \uparrow$	0↔1	1↔2	0↔2	E	M	H	$\eta^2 \downarrow$
ZipDepth [‡]	ECCV’26	D	—	0.56	0.58	0.51	0.60	0.51	0.09	0.60	0.42	0.37	0.54	0.63	0.59	0.46	0.081
HyDen [‡]	ICLR’26	D	—	0.44	0.42	0.45	0.45	0.42	0.03	0.79	0.73	0.50	0.64	0.76	0.78	0.70	0.103
FE2E	CVPR’26	D	—	0.68	0.70	0.65	0.68	0.65	0.05	0.59	0.39	0.41	0.49	0.62	0.55	0.49	0.047
DA3 Metric-L [24]	ICLR’26	D,P,N,GS	0.35 B	0.72	0.73	0.70	0.73	0.70	0.03	0.53	0.23	0.28	0.43	0.49	0.54	0.46	0.122
UniDepth V2 [27]	TPAMI’25	D	—	0.71	0.70	0.69	0.72	0.69	0.03	0.56	0.31	0.30	0.53	0.56	0.55	0.48	0.087
MoGe-2 ViT-L [‡] [29]	NeurIPS’25	D,P	0.33 B	0.52	0.48	0.54	0.53	0.48	0.06	0.66	0.43	0.51	0.51	0.66	0.66	0.61	0.116
Lotus-2 [30]	ICLR’25	D	—	0.36	0.35	0.38	0.35	0.35	0.03	0.64	0.34	0.66	0.44	0.64	0.61	0.55	0.082
Depth Pro [26]	ICLR’25	D	—	0.63	0.62	0.64	0.63	0.62	0.02	0.57	0.30	0.34	0.56	0.58	0.54	0.54	0.070
Metric3D V2 [28]	TPAMI’24	D,N	—	0.72	0.72	0.72	0.71	0.71	0.01	0.68	0.44	0.50	0.56	0.63	0.58	0.57	0.108
DA-V2 Large [25]	NeurIPS’24	D	0.34 B	0.60	0.60	0.55	0.66	0.55	0.11	0.54	0.27	0.33	0.43	0.59	0.49	0.51	0.097

TABLE V: T2: Video depth ($K=6$: AbsRel, RMSE, SILog, δ_1 , TGM, TGSE). Same diagnostic format as Table IV. All models evaluated on 250-frame clips.

Model	Venue	Out	Params	Desirability $\mathcal{J} \uparrow$				Summary		Overall	Phase-transition $\Phi \uparrow$			Scene-tier $\Phi \uparrow$			Amplif.
				overall	clu	int	cln	$\mathcal{J}_{\min} \uparrow$	$\mathcal{J}_{\Delta} \downarrow$	$\Phi \uparrow$	0↔1	1↔2	0↔2	E	M	H	$\eta^2 \downarrow$
DA3 (video) [24]	ICLR’26	D,P,N,GS	0.35 B	0.72	0.78	0.64	0.74	0.64	0.15	0.48	0.23	0.26	0.46	0.46	0.42	0.37	0.167
Video-DA [37]	CVPR’25	D	0.34 B	0.33	0.35	0.35	0.30	0.30	0.05	0.54	0.21	0.55	0.32	0.57	0.49	0.44	0.124
ViGeo [32]	arXiv’25	D	—	0.64	0.63	0.63	0.67	0.63	0.04	0.54	0.23	0.42	0.35	0.53	0.54	0.48	0.141
VGGT [31]	CVPR’25	D,P,N	1.26 B	0.72	0.70	0.73	0.72	0.70	0.03	0.77	0.63	0.58	0.64	0.70	0.65	0.74	0.087
MonST3R [36]	ICLR’25	D,P	0.27 B	0.55	0.58	0.49	0.57	0.49	0.09	0.52	0.27	0.37	0.39	0.49	0.55	0.38	0.174
GemDepth [33]	arXiv’25	D	—	0.69	0.76	0.61	0.71	0.61	0.15	0.45	0.17	0.31	0.34	0.46	0.39	0.37	0.148
DepthCrafter [38]	CVPR’25	D	—	0.30	0.31	0.31	0.26	0.26	0.05	0.57	0.33	0.54	0.30	0.60	0.49	0.47	0.171
DVD v1.1 [34]	arXiv’25	D	—	0.62	0.69	0.55	0.62	0.55	0.14	0.47	0.19	0.31	0.39	0.42	0.45	0.35	0.154
ChronoDepth [35]	CVPR’25	D	—	0.54	0.59	0.49	0.55	0.49	0.09	0.54	0.28	0.49	0.33	0.54	0.54	0.42	0.159
RollingDepth [39]	arXiv’24	D	—	0.22	0.23	0.24	0.20	0.20	0.03	0.58	0.27	0.59	0.34	0.60	0.54	0.47	0.166

have moderate Φ (0.54–0.58) but very low $\mathcal{J}$ (≤ 0.33): phase-stable because uniformly inaccurate.

All 10 video depth models show 2–4 $\times$ AbsRel amplification at the near range (0.3–1.0 m) during Interaction—the depth range where grasping occurs (Fig. 3). For 7/10 models, TGM degrades *more* than AbsRel during Interaction (Wilcoxon, $p < 0.001$).

D. T3: Tracking

E. T4: Relative Camera-Pose Estimation

Every RCPE model achieves higher $\mathcal{J}$ in the clean phase than in clutter—a consistent directional effect (Table VII). Yet none of the 28 completed model $\times$ split cells reaches statistical significance ($p > 0.05$, Holm-corrected), and Φ ranges from 0.86 to 0.97. This contrasts sharply with depth, where *every* model is significant. Relative pose is estimated from point correspondences that are insensitive to clutter provided the

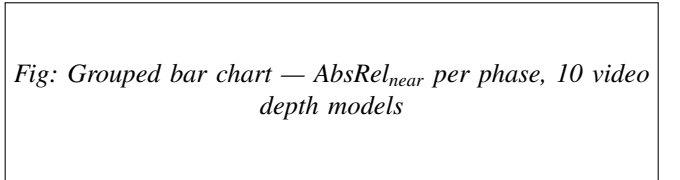


Fig. 3: Near-range (0.3–1.0 m) AbsRel across phases. Every video depth model degrades 2–4 $\times$ during Interaction.

object remains visible, whereas depth must assign metric values to every pixel—including occluders. DA3 leads on all completed splits (mean pose error 7.0°, $\mathcal{J}_{\min} = 0.38$), followed by Reloc3R. Translation-direction error dominates for every model (~ 7 – 28° vs. ~ 0.7 – 1.3° for rotation), identifying translation as the RCPE bottleneck. DUST3R achieves near-

TABLE VI: T3: Multi-object tracking ($K=4$: HOTA, DetA, AssA, LocA). Initialised with GT first-frame masks. Exo: D435 640×480, 3 phases. Ego: GoPro 1920×1080, Interaction only. Models span SAM-family VOS, memory-based VOS, detection-association MOT, and text-prompted pipelines. *Results in progress.*

Model	Venue	Out	Params	Desirability $\mathcal{J} \uparrow$					Summary		Overall	Phase-transition $\Phi \uparrow$			Scene-tier $\Phi \uparrow$			Amplif.
				overall	clu	int	cln	ego	$\mathcal{J}_{\min} \uparrow$	$\mathcal{J}_{\Delta} \downarrow$		$0 \leftrightarrow 1$	$1 \leftrightarrow 2$	$0 \leftrightarrow 2$	E	M	H	
SAM 3.1	arXiv'25	M	—															
SAM 2++	arXiv'25	M	—															
MOTIP	CVPR'25	B,ID	—															
SAMURAI [40]	NeurIPS'24	M	—															
SAM 2 [23]	arXiv'24	M	—															
MASA	CVPR'24	B,ID	—															
Grounded-SAM2	arXiv'24	M	—															
EdgeTAM [41]	CVPR'25	M	—															
Cutie [43]	CVPR'24	M	—															
DeAOT [42]	NeurIPS'22	M	—															

TABLE VII: T4: RCPE ($K=3$: AUC@5°, @10°, @20°). Exo only (D435 + T265 pose GT; GoPro lacks pose GT). Values averaged across difficulty splits (Easy/Medium/Hard). Φ and $\mathcal{J}$ compare clutter vs. clean (2 phases; single transition $0 \leftrightarrow 2$). [†]Easy split only.

Model	Venue	Out	Params	Desirability $\mathcal{J} \uparrow$			Summary		Overall $\Phi \uparrow$	Phase-transition 0↔2	Scene-tier $\Phi \uparrow$		
				overall	clu	cln	$\mathcal{J}_{\min} \uparrow$	$\mathcal{J}_{\Delta} \downarrow$			E	M	H
DA3 [24]	ICLR'26	D,P,N,GS	1.36 B	0.42	0.38	0.45	0.38	0.07	0.87	0.87	0.87	0.89	0.86
VGGT [31]	CVPR'25	D,P,N	1.26 B	0.37	0.34	0.41	0.34	0.06	0.91	0.91	0.96	0.94	0.85
Reloc3R [44]	CVPR'25	P	0.43 B	0.40	0.37	0.42	0.36	0.05	0.91	0.91	0.92	0.89	0.91
Pi3X [45]	arXiv'25	P	0.99 B	0.38	0.34	0.41	0.34	0.06	0.93	0.93	0.93	0.93	0.92
MonST3R [36]	ICLR'25	D,P	0.27 B	0.09	0.08	0.11	0.06	0.05	0.86	0.86	0.90	0.81	0.86
MUS3R [49]	CVPR'25	P	0.42 B	0.27	0.24	0.29	0.23	0.05	0.92	0.92	0.91	0.93	0.92
Fast3R [48]	CVPR'25	P	0.65 B	0.28	0.26	0.30	0.26	0.03	0.95	0.95	0.95	0.95	0.95
CUT3R [47]	CVPR'25	D,P	0.46 B	0.31	0.28	0.34	0.27	0.06	0.91	0.91	0.95	0.86	0.92
MASt3R [†] [46]	ECCV'24	P	0.69 B	0.33	0.30	0.35	0.30	0.05	0.97	0.97	0.97	—	—
DUST3R [50]	CVPR'24	P	0.57 B	0.18	0.16	0.19	0.16	0.02	0.97	0.97	0.95	0.99	0.96

Fig: Scatter plot — Φ vs $\mathcal{J}$, all models across tasks, colour by task

Fig. 4: Φ (invariance) vs. $\mathcal{J}$ (quality). Top-right = accurate and stable. Bottom-right = uniformly poor (high Φ by default). $\mathcal{J}_{\min}$ correctly ranks models; Φ alone does not.

perfect phase invariance ($\Phi = 0.97$) but the lowest $\mathcal{J}_{\min}$ after MonST3R (0.16)—the same uniformly-poor stability trap seen in depth.

F. T5–T7: Visual Question Answering

G. Rankings Change Across Tasks

Three foundation models—DA3, VGGT, and MonST3R—appear in both the video depth and RCPE evaluations. Their $\mathcal{J}_{\min}$ rankings diverge: VGGT ranks #1 on video depth ($\mathcal{J}_{\min} = 0.700$) but drops to #3 on RCPE ($\mathcal{J}_{\min} = 0.341$); DA3 moves from #2 on video depth to #1 on RCPE ($\mathcal{J}_{\min} = 0.380$). MonST3R is weak on both (#7 depth, #10 pose). A single “best foundation model” does not exist—the answer depends on which downstream task the pipeline serves.

H. Φ vs. $\mathcal{J}$: Two Axes, Not One

Three video depth models (RollingDepth, DepthCrafter, Video-DA) cluster in the high- Φ , low- $\mathcal{J}$ corner—phase-stable because uniformly inaccurate. DUST3R shows the same in RCPE: $\Phi = 0.97$ but $\mathcal{J}_{\min} = 0.16$. Within each task, Φ and $\mathcal{J}_{\min}$ are uncorrelated ($|r| < 0.15$), confirming that phase robustness and baseline quality are independent axes that must be reported jointly.

I. Image vs. Video Depth

Comparing the two depth tasks on overlapping accuracy metrics (AbsRel, RMSE, δ_1 , SILog), video models achieve comparable $\mathcal{J}_{\min}$ to image models (Mann–Whitney $p = 0.31$) but significantly lower Φ ($p = 0.021$): temporal context does not improve phase robustness. VGGT is the exception—it achieves the highest Φ on video depth (0.771) and the highest $\mathcal{J}_{\min}$ (0.700), likely because its multi-view transformer jointly reasons over the full clip. For 7/10 video models, temporal consistency (TGM) degrades *more* than spatial accuracy during Interaction (Wilcoxon, $p < 0.001$), revealing a failure mode invisible to frame-level benchmarks.

VI. DISCUSSION

Leaderboard rankings are unreliable for deployment. Metric3D V2 leads image depth ($\mathcal{J}_{\min} = 0.711$) yet has the highest Φ among metric models (0.675); it is the most accurate *and* the most phase-sensitive. Conversely, HyDen achieves the

TABLE VIII: T5–T7: VQA via 10 VLMs. All models output bounding boxes (native or prompted); IoU is the primary metric. T5: object recognition and attribute grounding. T6: 3D-grounded spatial reasoning. T7: multi-image in-context reasoning across phases. Exo: D435 640×480, 3 phases. Ego: GoPro 1920×1080, Interaction only. Bbox: `loc` = native localization token, `box` = native box coordinates, `P` = prompted. *Results in progress.*

Model	Venue	Out	Params	Desirability $\mathcal{J} \uparrow$					Summary		Overall	Phase-transition $\Phi \uparrow$			Scene-tier $\Phi \uparrow$			Amplif.
				overall	clu	int	cln	ego	$\mathcal{J}_{\min} \uparrow$	$\mathcal{J}_{\Delta} \downarrow$	$\Phi \uparrow$	0↔1	1↔2	0↔2	E	M	H	$\eta^2 \downarrow$
T5: General Awareness																		
Qwen2.5-VL [53]	arXiv'25	box	7B															
Qwen2.5-VL [53]	arXiv'25	box	3B															
LLaVA-OneVision [56]	arXiv'24	P	7B															
Gemma 4 [52]	arXiv'25	P	12B															
Gemma 4 [52]	arXiv'25	P	4B															
Phi-3.5-Vision [54]	arXiv'24	P	4.2B															
PaliGemma 2 [51]	arXiv'24	loc	10B															
PaliGemma 2 [51]	arXiv'24	loc	3B															
Idefics3 [55]	arXiv'24	P	8B															
InternVL 2.5 [57]	arXiv'24	P	8B															
T6: Spatial Awareness																		
Qwen2.5-VL [53]	arXiv'25	box	7B															
Qwen2.5-VL [53]	arXiv'25	box	3B															
LLaVA-OneVision [56]	arXiv'24	P	7B															
Gemma 4 [52]	arXiv'25	P	12B															
Gemma 4 [52]	arXiv'25	P	4B															
Phi-3.5-Vision [54]	arXiv'24	P	4.2B															
PaliGemma 2 [51]	arXiv'24	loc	10B															
PaliGemma 2 [51]	arXiv'24	loc	3B															
Idefics3 [55]	arXiv'24	P	8B															
InternVL 2.5 [57]	arXiv'24	P	8B															
T7: In-Context QA																		
Qwen2.5-VL [53]	arXiv'25	box	7B															
Qwen2.5-VL [53]	arXiv'25	box	3B															
LLaVA-OneVision [56]	arXiv'24	P	7B															
Gemma 4 [52]	arXiv'25	P	12B															
Gemma 4 [52]	arXiv'25	P	4B															
Phi-3.5-Vision [54]	arXiv'24	P	4.2B															
PaliGemma 2 [51]	arXiv'24	loc	10B															
PaliGemma 2 [51]	arXiv'24	loc	3B															
Idefics3 [55]	arXiv'24	P	8B															
InternVL 2.5 [57]	arXiv'24	P	8B															

best Φ (0.788) but the second-worst $\mathcal{J}_{\min}$ (0.421). Across tasks, VGGT leads video depth but drops to #3 on RCPE; DA3 is #1 on pose but #2 on depth. No single ranking generalizes.

Near-range depth is the canary metric. The 0.3–1.0 m range degrades 2–4× more than aggregate AbsRel during Interaction—precisely where grasping occurs. We recommend future depth benchmarks report range-stratified metrics.

Depth degrades; pose does not. The asymmetry between depth ($\Phi \leq 0.79$, all significant) and pose ($\Phi \geq 0.81$, none significant) suggests that dense per-pixel prediction is fundamentally more fragile to scene-state change than sparse geometric matching. This has practical implications: pipelines that can operate on pose alone may be more robust to manipulation-induced scene changes than those requiring per-frame metric depth.

VII. CONCLUSION

RPX evaluates pretrained perception under the conditions robots actually face: commodity RGB-D, scenes that change during manipulation, and multiple tasks on shared data. Across 29 depth and pose models, the central finding is a task-dependent asymmetry: scene-phase transitions significantly degrade every depth model ($p < 0.001$) but leave relative

pose estimation statistically unaffected ($p > 0.05$). Near-range depth collapse (2–4× at 0.3–1.0 m) and temporal flicker during interaction are robotics-critical failure modes invisible to existing benchmarks. No single foundation model dominates across tasks—VGGT leads on video depth, DA3 on both image depth and pose—motivating task-specific model selection rather than one-size-fits-all backbone deployment.

Future work. Closing the perception-to-manipulation loop—measuring how $\mathcal{J}_{\min}$ predicts downstream task success on real hardware—is the key next step.

Data, code, and the `rpx-benchmark` toolkit: <https://anonymous.4open.science/r/RPX>.

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